Pseudorandom generators for formulas of low-communication gates via polynomial fooling and NIH recycling

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Editorial note. Editable transcription of selected formally published CCBY3.0 Unported PRG results; nonexclusive nativev1 private. MainTheorem2 complete construction mechanisms and necessaryAppendixA.2 retained; GIP lowerbound/AppendixA.1 are shared uncounted source-local context, not another task. UnrelatedMCSP sketches removed, other gateclasses/parameters and directMKtP consequence grouped uncounted. Original author parameters/inequalities, including printed extractor bounds in prior-fact statement, preserved without mathematical correction or new assumptions. Wrapper uses correctedSet/ceil/floor dependency handling and independentpreamble copy, metadatafiveauthors unchanged. Noinput/include/assets. All production/validationYupeng; root owns compilation.

Allowed standard prerequisites (editorial).

- Low-degree formula approximation, communication rectangles, explicit extractors and hardness-versus-randomness hybrids. RequiredformalSection2; complete source-local polynomial fooling coefficient argument, NIH recursive construction andAppendixA.2 analysis retained; Section3/GIP proof shared uncounted context

Theorem 2 (Pseudorandom generators)**.** *Let $\mathcal{G}$ be a class of n-bits functions. Then,*

1. *In the context of parity functions, there is a* PRG *that ε-fools* $\mathsf{FORMULA}[s]\circ\mathsf{XOR}$ *of seed length*

$$\ell \;=\; O\big(\sqrt{s}\cdot\log(s)\cdot\log(1/\varepsilon)+\log(n)\big)\,.$$

2. *In the context of two-party randomized communication complexity, there is a* PRG *that ε-fools* $\mathsf{FORMULA}[s]\circ\mathcal{G}$ *of seed length*

$$\ell \;=\; n/2+O\Big(\sqrt{s}\cdot\Big(R^{(2)}_{\varepsilon/(6s)}(\mathcal{G})+\log(s)\Big)\cdot\log(1/\varepsilon)\Big)\,.$$

More generally, for every $k(n)\geq 2$, let $\mathcal{G}$ be the class of functions that have k-party number-in-hand (NIH) *$(\varepsilon/6s)$-error randomized communication protocols of cost at most $R^{(k\text{-}\mathsf{NIH})}_{\varepsilon/(6s)}$. There exists a* PRG *that ε-fools*

$\mathsf{FORMULA}[s] \circ \mathcal{G}$ *with seed length*

$$\ell \;=\; n/k + O\Big(\sqrt{s} \cdot \Big(R^{(k\text{-NIH})}_{\varepsilon/(6s)} + \log(s)\Big) \cdot \log(1/\varepsilon) + \log(k)\Big) \cdot \log(k).$$

3. *In the setting of k-party* NOF *randomized communication complexity, there is a* PRG *that ε-fools* $\mathsf{FORMULA}[s] \circ \mathcal{G}$ *of seed length*

$$\ell \;=\; n - \frac{n}{O\Big(\sqrt{s} \cdot k \cdot 4^k \cdot \Big(R^{(k)}_{\varepsilon/(2s)}(\mathcal{G}) + \log(n)\Big) \cdot \log(n/\varepsilon)\Big)}.$$

2 Preliminaries

2.1 Notation

Let $n \in \mathbb{N}$; we denote $\{1, \ldots, n\}$ by $[n]$, and denote by U_n the uniform distribution over $\{0,1\}^n$. We use $\widetilde{O}(\cdot)$ (and $\widetilde{\Omega}(\cdot)$) to hide polylogarithmic factors. That is, for any $f\colon \mathbb{N} \to \mathbb{N}$, we have that $\widetilde{O}(f(n)) = O(f(n) \cdot \mathrm{polylog}(f(n)))$.

In this paper, we will mainly use $\{-1,1\}$ as the Boolean basis. In some parts of this paper, we will use the $\{0,1\}$ basis for the simplicity of the presentation. This will be specified in corresponding sections.

2.2 De Morgan formulas and extensions

Definition 8. *An n-variate* de Morgan formula *is a directed rooted tree; its non-leaf vertices (henceforth,* internal gates*) take labels from* $\{\mathrm{AND}, \mathrm{OR}, \mathrm{NOT}\} = \{\wedge, \vee, \neg\}$ *and its leaves (henceforth,* variable gates*) take labels from the set of variables* $\{x_1, \ldots, x_n\}$*. Each internal gate has bounded in-degree (henceforth,* fan-in*); the* NOT *gate in particular has fan-in* 1 *and every variable gate has fan-in* 0*. The* size *of a de Morgan formula is the number of its leaf gates.*

In this work, we denote by $\mathsf{FORMULA}[s]$ the class of Boolean functions computable by size-s de Morgan formulas. Let $\mathcal{G}$ denote some class of Boolean functions; then, we denote by $\mathsf{FORMULA}[s] \circ \mathcal{G}$ the class of functions computable by some size-s de Morgan formula where its leaves are labelled by functions in $\mathcal{G}$.

2.3 Approximating polynomials

Definition 9 (Point-wise approximation)**.** *For a Boolean function $f\colon \{-1,1\}^n \to \{-1,1\}$, we say that the function $\tilde{f}\colon \{-1,1\}^n \to \mathbb{R}$ ε-approximates f if for every $z \in \{-1,1\}^n$,*

$$\Big|f(z) - \tilde{f}(z)\Big| \leq \varepsilon.$$

We will need the following powerful result for the approximating degree of de Morgan formulas.

Theorem 10 ([9], see also [3])**.** *Let $s > 0$ be an integer and $0 < \varepsilon < 1$. Any de Morgan formula $F\colon \{-1,1\}^n \to \{-1,1\}$ of size s has a ε-approximating polynomial of degree $d = O(\sqrt{s} \cdot \log(1/\varepsilon))$. That is, there exists a degree-d polynomial $p\colon \{-1,1\}^n \to \mathbb{R}$ over the reals such that for every $z \in \{-1,1\}^n$,*

$$|p(z) - F(z)| \leq \varepsilon.$$

Note that Theorem 10 still holds if we use $\{0,1\}$ as the Boolean basis.

2.4 Communication complexity

We use standard definitions from communication complexity. In this paper we consider the standard two party model of Yao and its generalizations to multiparty setting. We denote deterministic communication complexity of a Boolean function by $D(f)$ in the two party setting. We refer to [6] for standard definitions from communication complexity.

Definition 11. *Let $f\colon \{0,1\}^n \to \{0,1\}$ be a Boolean function. The* communication matrix of f*, namely M_f, is a $2^{n/2} \times 2^{n/2}$ matrix defined by $(M_f)_{x,y} := f(x,y)$.*

Definition 12. *A* rectangle *is a set of the form $A \times B$, for $A, B \subseteq \{0,1\}^n$. A* monochromatic rectangle *is a rectangle S such that for all pairs $(x,y) \in S$ the value $f(x,y)$ is the same.*

Lemma 13. *Let Π be a protocol that computes $f\colon \{0,1\}^n \to \{0,1\}$ with at most D bits of communication. Then, Π induces a partition of M_f into at most 2^D monochromatic rectangles.*

Given a protocol, its *transcript* is the sequence of bits communicated.

Lemma 14. *For every transcript z of some communication protocol, the set of inputs (x,y) that generate z is a rectangle.*

Below, we recount the definitions of two multiparty communication models used in this work, namely the number-on-forehead and the number-in-hand models.

Definition 15 ("Number-on-forehead" communication model; informal)**.** *In the k-party "number-on-forehead" communication model, there are k players and k strings $x_1, \ldots, x_k \in \{0,1\}^{n/k}$ and player i gets all the strings except for x_i. The players are interested in computing a value $f(x_1, \ldots, x_k)$, where $f\colon \{0,1\}^n \to \{0,1\}$ is some fixed function. We denote by $D^{(k)}(f)$ the number of bits that must be exchanged by the best possible number on forehead protocol solving f.*

We also use the following weaker communication model.

Definition 16 ("Number-in-hand" communication model; informal)**.** *In the k-party "number-in-hand" communication model, there are k players and k strings $x_1, \ldots, x_k \in \{0,1\}^{n/k}$ and player i gets only x_i. The players are interested in computing a value $f(x_1, \ldots, x_k)$, where $f\colon \{0,1\}^n \to \{0,1\}$ is some fixed function. We denote by $D^{(k\text{-NIH})}(f)$ the number of bits that must be exchanged by the best possible communication protocol.*

Note that $D^{(k\text{-NIH})}(f) \leq (1 - 1/k) \cdot n + 1$, for any n-variate Boolean function f, as if $k-1$ players write on the blackboard their string, then the player that did not reveal her input may compute $f(x_1, \ldots, x_k)$ on her own and then publish it.

For the communication models mentioned above, there are also bounded-error randomized versions, denoted by R_δ, $R_\delta^{(k)}$, and $R_\delta^{(k\text{-NIH})}$, respectively, where $0 < \delta < 1$ is an upper bound on the error probability of the protocol. In this setting, the players have access to some shared random string, say

r, and the aforementioned error probability of the protocol is considered over the possible choices of r. Moreover, we require the error to be at most δ on each fixed choice of inputs.

We can extend the definitions of the communication complexity measures, defined above, to classes of Boolean functions, in a natural way. That is, for any communication complexity measure $M \in \left\{D, D^{(k)}, D^{(k\text{-NIH})}, R_\delta, R_\delta^{(k)}, R_\delta^{(k\text{-NIH})}\right\}$ and for any class of Boolean functions $\mathcal{G}$, we may define

$$M(\mathcal{G}) := \max_{g \in \mathcal{G}} M(g)\,.$$

We note that throughout this paper, we denote by n the number of input bits for the function regardless the communication models. In the k-party communication setting (either NOF or NIH), we assume without loss of generality that n is divisible by k.

2.5 Pseudorandomness

A PRG against a class of functions $\mathfrak{C}$ is a deterministic procedure G mapping short Boolean strings (seeds) to longer Boolean strings, so that G's output "looks random" to every function in $\mathfrak{C}$.

Definition 17 (Pseudorandom generators)**.** *Let $G\colon \{-1,1\}^\ell \to \{-1,1\}^n$ be a function, $\mathfrak{C}$ be a class of Boolean functions, and $0 < \varepsilon < 1$. We say that G is a* pseudorandom generator of seed length ℓ that ε-fools $\mathfrak{C}$ *if, for every function $f \in \mathfrak{C}$, it is the case that*

$$\left| \mathop{\mathbf{E}}_{z \sim \{-1,1\}^\ell}[f(G(z))] - \mathop{\mathbf{E}}_{x \sim \{-1,1\}^n}[f(x)] \right| \leq \varepsilon.$$

A PRG G outputting n bits is called *explicit* if G can be computed in $\mathrm{poly}(n)$ time. All PRGs stated in this paper are explicit.

GIP_n^k. The Generalized Inner Product function $\mathsf{GIP}_n^k\colon \{0,1\}^n \to \{0,1\}$ is defined as

$$\mathsf{GIP}_n^k\left(x^{(1)}, x^{(2)}, \ldots, x^{(k)}\right) = \sum_{j=1}^{n/k} \bigwedge_{i=1}^{k} x_j^{(i)} \pmod 2,$$

where $x^{(i)} \in \{0,1\}^{n/k}$ for each $i \in [k]$.

3 Lower bounds

In this section, we prove an average-case lower bound for the generalized inner product function against $\mathsf{FORMULA} \circ \mathcal{G}$, where $\mathcal{G}$ is the set of functions that have low-cost randomized communication protocols in the number-on-forehead setting. This corresponds to Item 1 of Theorem 1 of the original publication. Items 2 and 3 rely on our PRG constructions, and the proofs are deferred to Section 4 of the original publication.

Theorem 19. *For any integer $k \geq 2$, $s > 0$ and any class of functions $\mathcal{G}$, let $C\colon \{-1,1\}^n \to \{-1,1\}$ be a function in $\mathsf{FORMULA}[s] \circ \mathcal{G}$ such that*

$$\mathop{\mathbf{Pr}}_{x\sim\{-1,1\}^n}\left[C(x) = \mathsf{GIP}_n^k(x)\right] \geq 1/2 + \varepsilon.$$

Then

$$s = \Omega\left(\frac{n^2}{k^2 \cdot 16^k \cdot \left(R^{(k)}_{\varepsilon/(2n^2)}(\mathcal{G}) + \log n\right)^2 \cdot \log^2(1/\varepsilon)}\right).$$

We need a couple useful lemmas from [11], whose proofs are presented in Section A.1 (Theorem 50 and Theorem 51) for completeness.

Lemma 20 ([11])**.** *Let $\mathcal{D}$ be a distribution over $\{-1,1\}^n$, and let $f, C\colon \{-1,1\}^n \to \{-1,1\}$ be such that*

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x) = f(x)] \geq 1/2 + \varepsilon.$$

Let $\tilde{C}\colon \{-1,1\}^n \to \mathbb{R}$ be a ε-approximating function of C, i.e., for every $x \in \{-1,1\}^n$, $|C(x) - \tilde{C}(x)| \leq \varepsilon$. Then,

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x) \cdot f(x)] \geq \varepsilon.$$

Lemma 21 ([11])**.** *Let $\mathcal{D}$ be a distribution over $\{-1,1\}^n$ and let $\mathcal{G}$ be a class of functions. For $f\colon \{-1,1\}^n \to \{-1,1\}$, suppose that $D\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{FORMULA}[s] \circ \mathcal{G}$ is such that*

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[D(x) = f(x)] \geq 1/2 + \varepsilon_0.$$

Then there exists some $h\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{XOR}_{O\left(\sqrt{s}\cdot\log(1/\varepsilon_0)\right)} \circ \mathcal{G}$ such that

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[h(x) \cdot f(x)] \geq \frac{1}{s^{O\left(\sqrt{s}\cdot\log(1/\varepsilon_0)\right)}}.$$

We also need the following communication-complexity lower bound for GIP.

Theorem 22 ([4, Theorem 2])**.** *For any $k \geq 2$, any function that computes GIP_n^k on more than $1/2+\delta$ fraction of the inputs (over uniformly random inputs) must have k-party deterministic communication complexity at least $\Omega\big(n/(k \cdot 4^k) - \log(1/\delta)\big)$.*

We first show that if a function correlates with some small formula, whose leaves are functions with low *randomized* communication complexity, then it also correlates non-trivially with some function of relatively low *deterministic* communication complexity.

Lemma 23. *For any distribution $\mathcal{D}$ over $\{-1,1\}^n$, and any class of functions $\mathcal{G}$, let $f\colon \{-1,1\}^n \to \{-1,1\}$ and $C\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{FORMULA}[s] \circ \mathcal{G}$ be such that*

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x) = f(x)] \geq 1/2 + \varepsilon.$$

Then there exists a function h, with k-party deterministic communication complexity at most

$$O\left(R^{(k)}_{\varepsilon/(2s)}(\mathcal{G}) \cdot \sqrt{s} \cdot \log(1/\varepsilon)\right),$$

such that

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[h(x) = f(x)] \geq 1/2 + 1/s^{O(\sqrt{s}\cdot\log(1/\varepsilon))}.$$

Proof. Let $C = F(g_1, g_2 \ldots, g_s)$ be the function in $\mathsf{FORMULA}[s] \circ \mathcal{G}$, where F is a formula and $g_1, g_2, \ldots, g_s$ are leaf functions from the class $\mathcal{G}$. For each g_i, consider a k-party randomized protocol Π_i of cost at most $R = R^{(k)}_{\varepsilon/(2s)}(\mathcal{G})$ that has an error $\varepsilon/(2s)$. Now consider the following function

$$\tilde{C}(x) := \mathop{\mathbf{E}}_{\Pi_1,\Pi_2,\ldots,\Pi_s}[D(x)],$$

where

$$D(x) := F(\Pi_1(x), \Pi_2(x), \ldots, \Pi_s(x)).$$

Note that for any fixed choice of $(\Pi_1, \Pi_2, \ldots, \Pi_s)$, D is a formula whose leaves are functions with *deterministic* communication complexity at most R. Next, we show the following.

Claim 24. *The function $\tilde{C}$ ε-approximates C.*

Proof of Theorem 24. First note that since each Π_i is a $(\varepsilon/(2s))$-error randomized protocol, by taking the union bound over the s leaf functions, we have that for every input $x \in \{-1, 1\}^n$,

$$\mathop{\mathbf{Pr}}_{\Pi_1,\Pi_2,\ldots,\Pi_s}[\Pi_1(x) = g_1(x) \wedge \Pi_2(x) = g_2(x) \wedge \cdots \wedge \Pi_s(x) = g_s(x)] \geq 1 - \varepsilon/2.$$

Denote by $\mathcal{E}$ the event $\Pi_1(x) = g_1(x) \wedge \Pi_2(x) = g_2(x) \wedge \cdots \wedge \Pi_s(x) = g_s(x)$. We have for every $x \in \{-1, 1\}^n$,

$$\begin{aligned}
\tilde{C}(x) &= \mathop{\mathbf{E}}_{\Pi_1,\Pi_2,\ldots,\Pi_s}[D(x)] \\
&= \mathbf{E}[D(x) \mid \mathcal{E}] \cdot \mathbf{Pr}[\mathcal{E}] + \mathbf{E}[D(x) \mid \neg\mathcal{E}] \cdot \mathbf{Pr}[\neg\mathcal{E}] \\
&= C(x) \cdot \mathbf{Pr}[\mathcal{E}] + \mathbf{E}[D(x) \mid \neg\mathcal{E}] \cdot \mathbf{Pr}[\neg\mathcal{E}].
\end{aligned}$$

On the one hand, we have

$$\tilde{C}(x) = C(x) \cdot \mathbf{Pr}[\mathcal{E}] + \mathbf{E}[D(x) \mid \neg\mathcal{E}] \cdot \mathbf{Pr}[\neg\mathcal{E}] \leq C(x) + \varepsilon/2.$$

On the other hand, we get

$$\tilde{C}(x) = C(x) \cdot \mathbf{Pr}[\mathcal{E}] + \mathbf{E}[D(x) \mid \neg\mathcal{E}] \cdot \mathbf{Pr}[\neg\mathcal{E}] \geq C(x) \cdot (1 - \varepsilon/2) + (-1) \cdot (\varepsilon/2) \geq C(x) - \varepsilon.$$

This completes the proof of the claim. □

Now by Theorem 24 and Theorem 20, we have

$$\mathop{\mathbf{E}}_{x \sim \mathcal{D}}[\tilde{C}(x) \cdot f(x)] \geq \varepsilon. \tag{1}$$

By the definition of $\tilde{C}$, Equation (1) implies that there exists some D, which is a formula whose leaves are functions with *deterministic* communication complexity at most R, such that

$$\mathop{\mathbf{E}}_{x \sim \mathcal{D}}[D(x) \cdot f(x)] \geq \varepsilon,$$

which implies

$$\mathop{\mathbf{Pr}}_{x \sim \mathcal{D}}[D(x) = f(x)] \geq 1/2 + \varepsilon/2.$$

Then by Theorem 21, there exists a function h, which can be expressed as the XOR of at most $O(\sqrt{s} \cdot \log(1/\varepsilon))$ leaf functions in D, such that

$$\mathop{\mathbf{E}}_{x \sim \mathcal{D}}[h(x) \cdot f(x)] \geq \frac{1}{s^{O(\sqrt{s} \cdot \log(1/\varepsilon))}},$$

which again implies

$$\mathop{\mathbf{Pr}}_{x \sim \mathcal{D}}[h(x) = f(x)] \geq \frac{1}{2} + \frac{1}{s^{O\left(\sqrt{s} \cdot \log(1/\varepsilon)\right)}}.$$

Finally, note that the k-party deterministic communication complexity of h is at most

$$O(R \cdot \sqrt{s} \cdot \log(1/\varepsilon)),$$

where $R = R^{(k)}_{\varepsilon/(2s)}(\mathcal{G})$. □

We are now ready to show Theorem 19.

Proof of Theorem 19. Consider Theorem 23 with f being GIP^k_n and $\mathcal{D}$ being the uniform distribution. Consider Theorem 22 with $\delta = 1/s^{O(\sqrt{s} \cdot \log(1/\varepsilon))}$. We have

$$O\left(R^{(k)}_{\varepsilon/(2s)}(\mathcal{G}) \cdot \sqrt{s} \cdot \log(1/\varepsilon)\right) \geq n/(k4^k) - O\left(\sqrt{s} \cdot \log(s) \cdot \log(1/\varepsilon))\right),$$

which implies

$$s \geq \Omega\left(\frac{n^2}{k^2 \cdot 16^k \cdot \left(R^{(k)}_{\varepsilon/(2n^2)}(\mathcal{G}) + \log n\right)^2 \cdot \log^2(1/\varepsilon)}\right).$$ □

4 Pseudorandom generators

Some of our PRGs are obtained from a general framework that allows us to reduce the task of fooling $\mathsf{FORMULA} \circ \mathcal{G}$ to the task of fooling the class of functions which are the parity or conjunction of few functions from $\mathcal{G}$.

4.1 The general framework

We show that in order to get a PRG for the class of subquadratic-size formulas with leaf gates in $\mathcal{G}$, it suffices to get a PRG for very simple sublinear-size formulas: either $\mathsf{XOR} \circ \mathcal{G}$ or $\mathsf{AND} \circ \mathcal{G}$.

Theorem 25 (PRG for $\mathsf{FORMULA} \circ \mathcal{G}$ from PRG for $\mathsf{XOR} \circ \mathcal{G}$ or $\mathsf{AND} \circ \mathcal{G}$)**.** *Let $\mathcal{G}$ be a class of gates on n bits. For any integer $s > 0$ and any $0 < \varepsilon < 1$, there exists a constant $c > 0$ such that the following holds. If a distribution $\mathcal{D}$ over $\{-1,1\}^n$ $\left(2^{-c \cdot \sqrt{s} \cdot \log(s) \cdot \log(1/\varepsilon)}\right)$-fools the XOR (parity) or the AND (conjunction) of $c \cdot \sqrt{s} \cdot \log(1/\varepsilon)$ arbitrary functions from $\mathcal{G}$, then $\mathcal{D}$ also ε-fools $\mathsf{FORMULA}[s] \circ \mathcal{G}$.*

Proof. We first show the case where $\mathcal{D}$ fools the parity of a few functions from $\mathcal{G}$. The proof can be easily adapted to the case of conjunction.

Let $C = F(g_1, g_2 \ldots, g_s)$ be a function in $\mathsf{FORMULA}[s] \circ \mathcal{G}$, where F is a formula, and $g_1, g_2, \ldots, g_s$ are functions from the class $\mathcal{G}$. Let U be the uniform distribution over $\{-1, 1\}^n$. We need to show

$$\mathbf{E}[C(\mathcal{D})] \overset{\varepsilon}{\approx} \mathbf{E}[C(U)]. \tag{2}$$

Let p be a $(\varepsilon/3)$-approximating polynomial for F given by Theorem 10. Note that the degree of p is

$$d = O(\sqrt{s} \cdot \log(1/\varepsilon)).$$

Let us replace F, the formula part of C, with p and let

$$\tilde{C} := p(g_1, g_2 \ldots, g_s).$$

Since $\tilde{C}$ point-wisely approximates C, we have

$$\mathbf{E}[\tilde{C}(U)] \overset{\varepsilon/3}{\approx} \mathbf{E}[C(U)],$$

and

$$\mathbf{E}[\tilde{C}(\mathcal{D})] \overset{\varepsilon/3}{\approx} \mathbf{E}[C(\mathcal{D})].$$

Then to show Equation (2), it suffices to show

$$\mathbf{E}[\tilde{C}(\mathcal{D})] \overset{\varepsilon/3}{\approx} \mathbf{E}[\tilde{C}(U)].$$

We have

$$\begin{aligned}
\mathop{\mathbf{E}}_{x \sim \mathcal{D}}[\tilde{C}(x)] &= \mathop{\mathbf{E}}_{x \sim D}\left[\sum_{\substack{S \subseteq [s]: \\ |S| \leq d}} \hat{p}(S) \cdot \prod_{i \in S} g_i(x)\right] \\
&= \sum_{\substack{S \subseteq [s]: \\ |S| \leq d}} \hat{p}(S) \cdot \mathop{\mathbf{E}}_{x \sim \mathcal{D}}\left[\prod_{i \in S} g_i(x)\right]. \qquad (3)
\end{aligned}$$

Now note that for each $S \subseteq [s]$, $\prod_{i \in S} g_i(x)$ computes the XOR of at most d functions from $\mathcal{G}$. Using the fact the distribution $\mathcal{D}$ $\left(\delta = 1/2^{c \cdot \sqrt{s} \cdot \log(s) \cdot \log(1/\varepsilon)}\right)$-fools the XOR of any d functions from $\mathcal{G}$, we get

$$\begin{aligned}
\mathop{\mathbf{E}}_{x \sim \mathcal{D}}[\tilde{C}(x)] &= \sum_{\substack{S \subseteq [s]: \\ |S| \leq d}} \hat{p}(S) \cdot \mathop{\mathbf{E}}_{x \sim D}\left[\prod_{i \in S} g_i(x)\right] \\
&= \sum_{\substack{S \subseteq [s]: \\ |S| \leq d}} \hat{p}(S) \cdot \left(\mathop{\mathbf{E}}_{x \sim U}\left[\prod_{i \in S} g_i(x)\right] + \delta_S\right) \qquad \text{(where } |\delta_S| \leq \delta\text{)} \\
&= \sum_{\substack{S \subseteq [s]: \\ |S| \leq d}} \left(\hat{p}(S) \cdot \mathop{\mathbf{E}}_{x \sim U}\left[\prod_{i \in S} g_i(x)\right] + \hat{p}(S) \cdot \delta_S\right)
\end{aligned}$$

$$= \sum_{\substack{S\subseteq[s]:\\ |S|\le d}} \hat{p}(S)\cdot \mathop{\mathbf{E}}_{x\sim U}\left[\prod_{i\in S} g_i(x)\right] + \sum_{\substack{S\subseteq[s]:\\ |S|\le d}} \hat{p}(S)\cdot\delta_S$$
$$= \mathop{\mathbf{E}}_{x\sim U}[\tilde{C}(x)] + \sum_{\substack{S\subseteq[s]:\\ |S|\le d}} \hat{p}(S)\cdot\delta_S.$$

It remains to show

$$\left|\sum_{\substack{S\subseteq[s]:\\ |S|\le d}} \hat{p}(S)\cdot\delta_S\right| \le \varepsilon/3.$$

Note that because $p(z)\in[1-\varepsilon/3, 1+\varepsilon/3]$ for every $z\in\{-1,1\}^s$, we have

$$|\hat{p}(S)| = \left|\mathop{\mathbf{E}}_{z\sim\{-1,1\}^s}\left[p(z)\cdot\prod_{i\in S} z_i\right]\right| \le 1+\varepsilon/3 < 2.$$

Then,

$$\left|\sum_{\substack{S\subseteq[s]:\\ |S|\le d}} \hat{p}(S)\cdot\delta_S\right| \le \sum_{\substack{S\subseteq[s]:\\ |S|\le d}} |\hat{p}(S)|\cdot|\delta_S| \le \delta\cdot\sum_{\substack{S\subseteq[s]:\\ |S|\le d}} |\hat{p}(S)| \le \delta\cdot s^{O(\sqrt{s}\cdot\log(1/\varepsilon))} \le \varepsilon/3,$$

where the last inequality holds for some sufficiently large constant c.

To show the case of conjunction, we can write the approximating polynomial as the sum of all degree-d monomials, each of which is the AND of at most d variables. One way to do this is to use the domain $\{0,1\}$ instead of $\{-1,1\}$ in the above argument. We need to show that the coefficients in this case still have small magnitude.

Claim 26. *Let $p\colon \{-1,1\}^n\to\mathbb{R}$ be a degree-d polynomial of the form*

$$p(x) = \sum_{\substack{S\subseteq[n]:\\ |S|\le d}} \hat{p}(S)\cdot\prod_{i\in S} x_i,$$

and let $q\colon \{0,1\}^n\to\mathbb{R}$ be the corresponding polynomial of p over the domain $\{0,1\}^n$, of the form

$$q(y) = \sum_{\substack{T\subseteq[n]:\\ |T|\le d}} \hat{q}(T)\cdot\prod_{i\in T} y_i.$$

Then,

$$|q|_1 = \sum_{\substack{T\subseteq[n]:\\ |T|\le d}} |\hat{q}(T)| \le n^{O(d)}\cdot\max_{\substack{S\subseteq[n]:\\ |S|\le d}} |\hat{p}(S)|.$$

Proof. We have

$$q(y_1,y_2,\ldots,y_n) = p(1-2y_1, 1-2y_2,\ldots,1-2y_n)$$

$$= \sum_{\substack{S\subseteq[n]:\\ |S|\le d}} \hat{p}(S)\cdot\prod_{i\in S}(1-2y_i)$$

$$= \sum_{\substack{S\subseteq[n]:\\ |S|\le d}} \hat{p}(S)\cdot\left(\sum_{\ell\in\{0,1\}^{|S|}}\prod_{\substack{j\in S:\\ \ell_j=1}}-2y_j\right)$$

$$= \sum_{\substack{S\subseteq[n]:\\ |S|\le d}}\sum_{\ell\in\{0,1\}^{|S|}} \hat{p}(S)\cdot(-2)^{|\ell|}\cdot\prod_{\substack{j\in S:\\ \ell_j=1}} y_j. \quad (\text{where } |\ell|=\textstyle\sum_{i=1}^{|S|}\ell_i)$$

For a pair (S,ℓ) where $S\subseteq[n]$, $|S|\le d$ and $\ell\in\{0,1\}^{|S|}$, let us define the polynomial $q_{(S,\ell)}$ as

$$q_{(S,\ell)}(y) = \hat{p}(S)\cdot(-2)^{|\ell|}\cdot\prod_{\substack{j\in S:\\ \ell_j=1}} y_j.$$

Note that there are at most $n^d\cdot 2^d$ many pairs of such (S,ℓ)'s and for each (S,ℓ), we have

$$|q_{(S,\ell)}|_1 = \left|\hat{p}(S)\cdot(-2)^{|\ell|}\right| \le 2^d\cdot|\hat{p}(S)|.$$

Finally we have

$$|q|_1 = \left|\sum_{(S,\ell)} q_{(S,\ell)}\right|_1 \le \sum_{(S,\ell)}|q_{(S,\ell)}|_1 \le n^d\cdot 2^d\cdot 2^d\cdot\max_{\substack{S\subseteq[n]:\\ |S|\le d}}|\hat{p}(S)|,$$

as desired. □ (Theorem 26)

This completes the proof of Theorem 25. □ (Theorem 25)

4.2 Formulas of low-communication functions in the number-in-hand setting

In this subsection, we will use $\{0,1\}$ as the Boolean basis.

Theorem 27. *For any integers $k\ge 2$, $s>0$ and any $0<\varepsilon<1$, let $\mathcal{G}$ be the class of functions that have k-party number-in-hand $(\varepsilon/6s)$-error randomized communication protocols of cost at most R. There exists a PRG that ε-fools $\mathsf{FORMULA}[s]\circ\mathcal{G}$ with seed length*

$$n/k + O\left(\sqrt{s}\cdot(R+\log(s))\cdot\log(1/\varepsilon)+\log(k)\right)\cdot\log(k).$$

We need the following PRG that fools single functions with low communication complexity in the number-in-hand model. The proof is presented in Section A.2 (Theorem 52) for completeness.

Theorem 28 ([2, 5])**.** *For any $k\ge 2$, there exists a PRG that δ-fools any n-bits functions with k-party number-in-hand deterministic communication complexity of at most D', with seed length*

$$n/k + O\left(D'+\log(1/\delta)+\log(k)\right)\cdot\log(k).$$

Next, we show a PRG for $\mathsf{FORMULA} \circ \mathcal{G}$, where $\mathcal{G}$ is the class of functions with low-cost communication protocols in the number-in-hand setting. We first show for the case of deterministic protocols.

Theorem 29. *For any integers $k \geq 2$ and $s > 0$, let $\mathcal{G}$ be the class of functions whose k-party number-in-hand deterministic communication complexity are at most D. There is a PRG that ε-fools $\mathsf{FORMULA}[s] \circ \mathcal{G}$ of size s with seed length*

$$n/k + O\left(\sqrt{s} \cdot \log(1/\varepsilon) \cdot (D + \log(s)) + \log(k)\right) \cdot \log(k).$$

Proof. By Theorem 25, it suffices to show a PRG that $\left(\delta = 1/2^{c \cdot \sqrt{s} \cdot \log(s) \cdot \log(1/\varepsilon)}\right)$-fools every function that is the XOR of $t = c \cdot \sqrt{s} \cdot \log(1/\varepsilon)$ arbitrary functions from $\mathcal{G}$. Note that such a function has deterministic communication complexity at most $D' = t \cdot D$. Then Theorem 29 follows from Theorem 28. □

We now establish the randomized case.

Proof of Theorem 27. Let C be a function in $\mathsf{FORMULA}[s] \circ \mathcal{G}$. For each of the leaf functions in C, consider a k-party number-in-hand randomized protocol of cost at most R that has an error at most $\varepsilon/(6s)$. By taking a union bound over the s leaf functions and by viewing a randomized protocol as a distribution of deterministic protocols (as shown in the proof of Theorem 24), we get the following which is a (point-wisely) $(\varepsilon/3)$-approximating function for C:

$$\tilde{C}(x) := \sum_i p_i \cdot D_i(x),$$

where each $p_i \in [0, 1]$ is some probability density value (so $\sum_i p_i = 1$), and each D_i is a formula whose leaves are functions with *deterministic* communication complexity at most R. Then to ε-fool C, it suffices to $(\varepsilon/3)$-fool its $(\varepsilon/3)$-approximating function $\tilde{C}$. Also, since $\tilde{C}$ is a convex combination of the D_i's, it suffices to $(\varepsilon/3)$-fools all the D_i's. We will do this using the PRG form Theorem 29. We get that there exists a PRG that $(\varepsilon/3)$-fools each D_i with seed length

$$n/k + O\left(\sqrt{s} \cdot (R + \log(s)) \cdot \log(1/\varepsilon) + \log(k)\right) \cdot \log(k),$$

as desired. □

4.3 Applications: Fooling formulas of SYMs, LTFs, XORs, and $\mathbf{AC}^0$ circuits

4.3.1 $\mathsf{FORMULA} \circ \mathsf{SYM}$ and $\mathsf{FORMULA} \circ \mathsf{LTF}$

Here, we show how the PRG in Theorem 27 implies PRGs for $\mathsf{FORMULA} \circ \mathsf{LTF}$ and $\mathsf{FORMULA} \circ \mathsf{SYM}$.

Theorem 30. *For any size $s > 0$ and $0 < \varepsilon < 1$, there exists a PRG that ε-fools $\mathsf{FORMULA}[s] \circ \mathsf{LTF}$ with seed length*

$$O\left(n^{1/2} \cdot s^{1/4} \cdot \log(n) \cdot \log(n/\varepsilon)\right).$$

For $\mathsf{FORMULA}[s] \circ \mathsf{SYM}$, the seed length is

$$O\left(n^{1/2} \cdot s^{1/4} \cdot \log(n) \cdot \log(1/\varepsilon)\right).$$

We need the fact that the class of LTF has low communication complexity in the number-in-hand model. Consider the following k-party SUM-GREATER$_m$ problem where the i-th party holds a m-bit number z_i in hand and they want to determine whether $\sum_{i=1}^{k} z_i > \theta$, where θ is a fixed number known to all the parties. Nisan [8] gave an efficient randomized protocol (with public randomness) for this problem.

Theorem 31 ([8][1])**.** *Let $m > 0$ be an integer. For any integer $2 \leq k \leq m^{O(1)}$, and any $0 < \delta < 1$, there exists a δ-error randomized protocol of cost $O(k \cdot \log(m) \cdot \log(m/\delta))$ for the k-party* SUM-GREATER$_m$ *problem.*

By Theorem 31 and the fact that every linear threshold function on n bits has a representation such that the weights are $O(n \log(n))$ integers [7], we get the following.

Corollary 32. *For every $k \geq 2$ and $0 < \delta < 1$, the k-party number-in-hand δ-error randomized communication complexity of* LTF *is $O(k \cdot \log(n) \cdot \log(n/\delta))$.*

Proof of Theorem 30. By Theorem 32 and Theorem 27, for *every* $k \geq 2$ we get a PRG for $\mathsf{FORMULA} \circ \mathsf{LTF}$ of seed length

$$n/k + O\left(\sqrt{s} \cdot k \cdot \log(n) \cdot \log(ns/\varepsilon) \cdot \log(1/\varepsilon) + \log(k)\right) \cdot \log(k).$$

By choosing

$$k = \frac{n^{1/2}}{s^{1/4} \cdot \log(n) \cdot \log(n/\varepsilon)},$$

the claimed seed length follows from a simple calculation.

For $\mathsf{FORMULA} \circ \mathsf{SYM}$, note that every n-bit symmetric function has a deterministic k-party number-in-hand communication protocol of cost at most $k \cdot \log(n)$. Then the rest can be shown using a similar argument as above (by choosing $k = n^{1/2}/\left(s^{1/4} \cdot \log(n)\right)$). □

4.3.2 $\mathsf{FORMULA} \circ \mathsf{XOR}$

For the case of $\mathsf{FORMULA} \circ \mathsf{XOR}$, we get a PRG with better seed length.

Theorem 33. *For any size $s > 0$ and $0 < \varepsilon < 1$, there exists a PRG that ε-fools* $\mathsf{FORMULA}[s] \circ \mathsf{XOR}$ *with seed length*

$$O\left(\sqrt{s} \cdot \log(s) \cdot \log(1/\varepsilon) + \log(n)\right).$$

Proof. By Theorem 25, to fool $\mathsf{FORMULA}[s] \circ \mathcal{G}$, it suffices to $\left(\delta = 1/2^{O\left(\sqrt{s} \cdot \log(s) \cdot \log(1/\varepsilon)\right)}\right)$-fool the XOR of a few functions from $\mathcal{G}$, where $\mathcal{G}$ in this case is the set of all XOR functions. Note that the XOR of any set of XOR functions simply computes some XOR function. Therefore, we can use small-bias distribution, which fools every XOR function, to fool $\mathsf{FORMULA}[s] \circ \mathsf{XOR}$. Finally, note that there are known constructions for δ-bias distributions that use $O(\log(n/\delta))$ random bits (see e.g. [1]). □

[1] Viola [13] gave a δ-error randomized protocol for the k-party SUM-GREATER$_m$ problem of cost $O(k \cdot \log(k) \cdot \log(m/\delta))$, which is better than Nisan's protocol when $k = m^{o(1)}$.

4.3.3 $\mathsf{FORMULA} \circ \mathsf{AC}^0$

Another application of Theorem 25 is to take $\mathcal{G}$ to be the set all functions that can be computed by small constant-depth circuits (AC^0). Note the state-of-the-art PRG against size-M depth-d AC^0 has a seed length of $\log^{d+O(1)}(Mn) \cdot \log(1/\varepsilon)$ [10]. Below, let $\mathsf{AC}^0_{d,M}$ denote the class of depth-d circuits of size at most M.

Theorem 35. *For any size $s, m > 0$ and $0 < \varepsilon < 1$, there exists a PRG that ε-fools $\mathsf{FORMULA} \circ \mathsf{AC}^0_{d,M}$ of size s with seed length*

$$\log^{d+O(1)}(Mn) \cdot \sqrt{s} \cdot \log(s) \cdot \log(1/\varepsilon).$$

4.4 Formulas of low number-on-forehead communication leaf gates

In this section, we show a PRG with mild seed length for formulas of functions with low *multi-party number-on-forehead* communication complexity.

Theorem 37. *Let $\mathcal{G}$ be a class of n-bits functions. For any size $s > 0$, there exists a PRG that ε-fools $\mathsf{FORMULA}[s] \circ \mathcal{G}$, with seed length*

$$n - \frac{n}{O\left(\sqrt{s} \cdot k \cdot 4^k \cdot \left(R^{(k)}_{\varepsilon/(2s)}(\mathcal{G}) + \log(n)\right) \cdot \log(n/\varepsilon)\right)}.$$

The PRG is constructed using the hardness vs. randomness paradigm.

4.4.1 Hardness based PRGs

We show how to construct the PRG using the average-case hardness result for formulas of functions with low multi-party communication complexity (Theorem 19). We start with some notations. For $x \in \{-1,1\}^m$ and an integer k such that k divides m, we consider a partition of x into k equal-sized consecutive blocks and write $x = x^{(1)}, x^{(2)}, \ldots, x^{(k)}$, where $x^{(i)} \in \{-1,1\}^{m/k}$ for each $i \in [k]$.

Lemma 38. *For any integers $m, t, k > 0$ such that k divides m, t, let $\mathcal{G}$ be a class of functions on $mt + t$ bits, and let $G\colon \{-1,1\}^{m \times t} \to \{-1,1\}^{mt+t}$ be*

$$\begin{aligned} &G(x_1, x_2, \ldots, x_t) \\ &= \left(x_1^{(i)}, x_2^{(i)}, \ldots, x_t^{(i)}, \mathsf{GIP}^k_m\left(x_{(i-1)\cdot(t/k)+1}\right), \mathsf{GIP}^k_m\left(x_{(i-1)\cdot(t/k)+2}\right), \ldots, \mathsf{GIP}^k_m\left(x_{i\cdot(t/k)+1}\right)\right)_{i\in[k]}, \end{aligned}$$

where $x_1, x_2, \ldots, x_t \in \{-1,1\}^m$. Then G is a PRG that $(t\cdot\varepsilon)$-fools $\mathsf{FORMULA} \circ \mathcal{G}$ of size

$$s = \Omega\left(\frac{m^2}{k^2 \cdot 16^k \cdot \left(R^{(k)}_{\varepsilon/(2m^2)}(\mathcal{G}) + \log m\right)^2 \cdot \log^2(1/\varepsilon)}\right).$$

Proof. The high level idea is as follows. We argue that if there is a $\mathsf{FORMULA} \circ \mathcal{G}$ of the claimed size that breaks the PRG, then there is a $\mathsf{FORMULA} \circ \mathcal{G}'$ of the same size that computes GIP on m bits, where $\mathcal{G}'$ has a k-party communication

complexity that is at most that of $\mathcal{G}$ *with respect to the* m*-bit input*, and hence contradicts the $\mathsf{FORMULA} \circ \mathcal{G}'$ complexity of the generalized inner product function. The resulting formula is obtained by fixing some input bits of the original $\mathsf{FORMULA} \circ \mathcal{G}$ which breaks the PRG.

We use a hybrid argument. First consider the distribution given by G, where we replace each $\mathsf{GIP}(x_j)$ ($j \in [t]$) with a uniformly random bit; let us denote those random bits as U_j for $j \in [t]$ (note that this is just the uniform distribution). Then for each $j \in [t]$, define H_j to be the distribution that we substitute back $\mathsf{GIP}(x_1), \mathsf{GIP}(x_2), \ldots, \mathsf{GIP}(x_j)$ for the corresponding uniform bits in the previous distribution.

For the sake of contradiction, suppose there exists a $\mathsf{FORMULA} \circ \mathcal{G}$ C of size s such that

$$|\mathbf{Pr}[C(H_t) = 1] - \mathbf{Pr}[F(H_0) = 1]| > t \cdot \varepsilon.$$

By the triangle inequality, there exists a $1 \leq j \leq k$ such that

$$|\mathbf{Pr}[C(H_j) = 1] - \mathbf{Pr}[C(H_{j-1}) = 1]| > \varepsilon.$$

Then by averaging, there exist some fixings of $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_t$ and $U_{j+1}, \ldots, U_t$ to C such that the above inequality still holds. Let us denote by C' the circuit obtained by C after such fixings and assume without loss of generality $(k-1)t/k \leq j \leq t$. Then we have

$$\left|\mathbf{Pr}\left[C'\left(x_j^{(1)}, x_j^{(2)}, \ldots, x_j^{(k)}, \mathsf{GIP}(x_j)\right) = 1\right] - \mathbf{Pr}\left[C'\left(x_j^{(1)}, x_j^{(2)}, \ldots, x_j^{(k)}, U_j\right) = 1\right]\right| > \varepsilon. \tag{4}$$

By a standard "unpredictability implies pseudorandomness" argument [14], we can show that there is some circuit C'', obtained from C' by fixing some value for the last bit, that computes the generalized inner product function on m bits with probability greater than $1/2 + \varepsilon$ over uniformly random inputs. Note that the size of C'' is the same as C' (hence also C) , and also C'' can be computed by some $\mathsf{FORMULA} \circ \mathcal{G}'$, where $R_\delta^{(k)}(\mathcal{G}') \leq R_\delta^{(k)}(\mathcal{G})$ for every δ. This contradicts hardness of GIP for such circuits (Theorem 19). □

We are now ready to prove Theorem 37.

Proof of Theorem 37. Consider Theorem 38. Let $n = mt + t$, and we have $m = \left(\frac{n}{t} - 1\right)$. Then Theorem 38 gives a PRG that ε-fools $\mathsf{FORMULA} \circ \mathcal{G}$ of size

$$\begin{aligned} s &= \Omega\left(\frac{m^2}{k^2 \cdot 16^k \cdot \left(R^{(k)}_{\varepsilon/(2m^2)}(\mathcal{G}) + \log m\right)^2 \cdot \log^2(t/\varepsilon)}\right) \\ &\geq \Omega\left(\left(\frac{n}{t}\right)^2 / \left(k^2 \cdot 16^k \cdot \left(R^{(k)}_{\varepsilon/(2n^2)}(\mathcal{G}) + \log n\right)^2 \cdot \log^2(n/\varepsilon)\right)\right), \end{aligned}$$

which yields

$$t \geq \Omega\left(\frac{n}{\sqrt{s} \cdot k \cdot 4^k \cdot \left(R^{(k)}_{\varepsilon/(2n^2)}(\mathcal{G}) + \log n\right) \cdot \log(n/\varepsilon)}\right).$$

Note that the seed length in this case is $n - t$. □

4.4.2 MKtP lower bounds

The PRG in Theorem 37 is sufficient to give an MKtP lower bound for formulas of functions with low multi-party communication complexity.

Theorem 39. *For any integer $s > 0$ and any class of N-bit function $\mathcal{G}$, if* MKtP *on N-bit can be computed by some function* $\mathsf{FORMULA}[s] \circ \mathcal{G}$*, then*

$$s = \frac{N^2}{k^2 \cdot 16^k \cdot R^{(k)}_{1/3}(\mathcal{G}) \cdot \mathrm{polylog}(N)}.$$

Proof. Let C be a function in $\mathsf{FORMULA} \circ \mathcal{G}$ of size less than

$$\frac{N^2}{k^2 \cdot 16^k \cdot R^{(k)}_{1/3}(\mathcal{G}) \cdot \log^c(N)}$$

where $c > 0$ is some sufficiently large constant. By Theorem 37, we have that there is a PRG that $(1/3)$-fools C and its seed length is

$$N - \mathrm{polylog}(N).$$

Also, since the PRG is polynomial-time computable, we get that for every seed, the output of the PRG has Kt complexity at most $\theta = N - \mathrm{polylog}(N)$. However, consider the MKtP function with a threshold parameter θ; this function is not fooled by such a PRG, since it accepts every output of the PRG and rejects a uniformly random string with high probability. □

A Proofs of useful lemmas

A.1 Useful lemmas for formulas

The proofs in this section are essentially the same as that of [11].

Lemma 50 ([11], Theorem 20 restated)**.** *Let $\mathcal{D}$ be a distribution over $\{-1,1\}^n$, and let $f, C\colon \{-1,1\}^n \to \{-1,1\}$ be such that*

$$\Pr_{x\sim\mathcal{D}}[C(x) = f(x)] \geq 1/2 + \varepsilon.$$

Let $\tilde{C}\colon \{-1,1\}^n \to \mathbb{R}$ be a ε-approximating function of C, i.e., for every $x \in \{-1,1\}^n$, $|C(x) - \tilde{C}(x)| \leq \varepsilon$. Then,

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x) \cdot f(x)] \geq \varepsilon.$$

Proof. Note that since $\tilde{C}$ ε-approximate C, we have for every $x \in \{-1,1\}^n$

$$\tilde{C} \cdot C(x) \geq 1 - \varepsilon,$$

and

$$\tilde{C} \cdot (1 - C(x)) \geq -1 - \varepsilon.$$

Then,

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x) \cdot f(x)] = \mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x) \cdot f(x) \mid C(x) = f(x)] \cdot \Pr_{x\sim\mathcal{D}}[C(x) = f(x)]$$

$$
\begin{aligned}
&\quad + \mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x)\cdot f(x) \mid C(x)\neq f(x)]\cdot \mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x)\neq f(x)]\\
&\geq (1-\varepsilon)\cdot \mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x)=f(x)] + (-1-\varepsilon)\cdot\left(1-\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x)=f(x)]\right)\\
&= 2\cdot \mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x)=f(x)] - 1 - \varepsilon\\
&\geq 2\cdot(1/2+\varepsilon) - 1 - \varepsilon \geq \varepsilon,
\end{aligned}
$$

as desired. □

Lemma 51 ([11], Theorem 21 restated)**.** *Let $\mathcal{D}$ be a distribution over $\{-1,1\}^n$ and let $\mathcal{G}$ be a class of functions. For $f\colon \{-1,1\}^n \to \{-1,1\}$, suppose that $D\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{FORMULA}[s]\circ\mathcal{G}$ is such that*

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[D(x)=f(x)] \geq 1/2+\varepsilon_0.$$

Then there exists some $h\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{XOR}_{O\left(\sqrt{s}\cdot\log(1/\varepsilon_0)\right)}\circ\mathcal{G}$ such that

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[h(x)\cdot f(x)] \geq \frac{1}{s^{O\left(\sqrt{s}\cdot\log(1/\varepsilon_0)\right)}}.$$

Proof. Let

$$D = F(g_1, g_2 \ldots, g_s)$$

be a device in $\mathsf{FORMULA}\circ\mathcal{G}$ where F is a formula and $g_1, g_2, \ldots, g_s$ are function from $\mathcal{G}$.

Let $p\colon \{-1,1\}^s \to \mathbb{R}$ be a ε_0-approximating polynomial for F of degree $d = O(\sqrt{s}\cdot\log(1/\varepsilon_0))$. Note that we can write

$$p(z) = \sum_{\substack{S\subseteq[s]:\\ |S|\leq d}} \hat{p}(S)\cdot\prod_{i\in S} z_i.$$

Also, for each $S\subseteq[s]$, we have

$$|\hat{p}(S)| = \left|\mathop{\mathbf{E}}_{z\in\{-1,1\}^s}[p(z)\cdot\prod_{i\in S} z_i]\right| \leq 1+\varepsilon_0.$$

Now let

$$\tilde{D} := p(g_1, g_2 \ldots, g_s).$$

Note that $\tilde{D}$ is a ε_0-approximating function for D. Therefore, by Theorem 50, we have

$$
\begin{aligned}
\varepsilon_0 &\leq \mathop{\mathbf{E}}_{x\sim\mathcal{D}}[D(x)\cdot f(x)]\\
&= \mathop{\mathbf{E}}_{x\sim\mathcal{D}}\left[\left(\sum_{\substack{S\subseteq[s]:\\ |S|\leq d}} \hat{p}(S)\cdot\prod_{i\in S} g_i\right)\cdot f(x)\right]\\
&= \sum_{\substack{S\subseteq[s]:\\ |S|\leq d}} \hat{p}(S)\cdot \mathop{\mathbf{E}}_{x\sim\mathcal{D}}\left[\prod_{i\in S} g_i\cdot f(x)\right]
\end{aligned}
$$

$$\leq \sum_{\substack{S\subseteq[s]:\\ |S|\leq d}} (1+\varepsilon_0)\cdot \left|\mathop{\mathbf{E}}_{x\sim\mathcal{D}}\left[\prod_{i\in S} g_i\cdot f(x)\right]\right|.$$

The above equation is the sum of at most $s^{O(d)}$ summands. Therefore, there exists some $S\subseteq[s]$ such that

$$\left|\mathop{\mathbf{E}}_{x\sim\mathcal{D}}\left[\prod_{i\in S} g_i\cdot f(x)\right]\right| \geq \frac{\varepsilon_0}{(1+\varepsilon_0)\cdot s^{O(d)}} \geq \frac{1}{s^{O\left(\sqrt{s}\cdot\log(1/\varepsilon_0)\right)}},$$

which implies that there exists some h, such that either $h=\prod_{i\in S} g_i$ or $h=-\prod_{i\in S} g_i$, and

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}\left[h(x)\cdot f(x)\right] \geq \frac{1}{s^{O\left(\sqrt{s}\cdot\log(1/\varepsilon_0)\right)}}.$$

Finally, note that such h can be expressed as the XOR of at most d functions from $\mathcal{G}$. □

A.2 PRG for low-communication functions in the number-in-hand setting

In this subsection, we show how to fool functions with low communication complexity in the number-in-hand model.

Theorem 52 ([2, 5], Theorem 28 restated)**.** *For any $k\geq 2$, there exists a PRG that δ-fools any n-bits functions with k-party number-in-hand deterministic communication complexity at most D', with seed length*

$$n/k + O\left(D' + \log(1/\delta) + \log(k)\right)\cdot\log(k).$$

The PRG in Theorem 28 is based on the PRG by Impagliazzo, Nisan and Wigderson [5] that is used to derandomize "network algorithms" and space-bounded computation. We will need to use randomness extractors, which we review below.

Definition 53 (Min-entropy)**.** *Let X be a random variable. The* min-entropy *of X, denoted by $H_\infty(X)$, is the largest real number k such that $\mathbf{Pr}[X=x]\leq 2^{-k}$ for every x in the range of X. If X is a distribution over $\{-1,1\}^{\aleph}$ with $H_\infty(X)\geq k$, then X is called a* $(\aleph,k)$-source*.*

Definition 54 (Extractors)**.** *A function* $\mathrm{Ext}\colon \{-1,1\}^{\aleph}\times\{-1,1\}^{d}\to\{-1,1\}^{m}$ *is an* (k,ε)-extractor *if, for any $(\aleph,k)$-source X, and any test $T\colon \{-1,1\}^{m}\to\{-1,1\}$, it is the case that*

$$\left|\mathbf{Pr}[T(\mathrm{Ext}(X,U_d)\,X)=1]-\mathbf{Pr}[T(U_m)=1]\right|\leq\varepsilon.$$

Theorem 55 ([12, Theorem 6.22])**.** *For any integer $m,\kappa>0$ and $0<\delta'<0$, there exists an explicit (κ,δ') extractor* $\mathrm{Ext}\colon \{0,1\}^{m}\times\{0,1\}^{d}\to\{0,1\}^{m}$ *with $d=O(m-k+\log(1/\delta'))$.*

We are now ready to show Theorem 28.

Proof of Theorem 28. We first describe the construction of the PRG. In fact, we will construct a sequence of PRGs $G_0, G_1, \ldots, G_{\log(k)}$. We begin by specifying the parameters of these PRGs. Let $t = \log(k)$, and let

$$d = O\left(D' + \log(1/\delta) + t\right).$$

For $i = 0, 1, \ldots, t$, let

- $r_0 = n/k$,
- $r_i = r_{i-1} + d$.

Note that we have $r_i = n/k + i \cdot d$. Also, let

$$\mathrm{Ext}_i \colon \{0,1\}^{r_i} \times \{0,1\}^d \to \{0,1\}^{r_i}$$

be a (κ_i, δ')-extractor from Theorem 55, where

$$\kappa_i = r_i - D' - 2t - \log(1/\delta)$$

and

$$\delta' = \delta / \left(3^t \cdot 2^{D'}\right).$$

Note that the seed length of the extractors is $d = O\left(D' + \log(1/\delta) + t\right)$. Finally, define $G_i \colon \{0,1\}^{r_i} \to \{0,1\}^{n/2^{t-i}}$ recursively as follows

- $G_0(a) = a$, where $a \in \{0,1\}^{n/k}$.
- $G_i(a, z) = G_{i-1}(a) \circ G_{i-1}(\mathrm{Ext}_{i-1}(a, z))$, where $a \in \{0,1\}^{r_{i-1}}$ and $z \in \{0,1\}^d$.

We will show that $G_t \colon \{0,1\}^{r_t = n/k + t \cdot d} \to \{0,1\}^n$ fools any functions f with k-party number-in-hand deterministic communication complexity at most D'. First, note that such f can be written as

$$f(x_1, x_2, \ldots, x_k) = \sum_{i=1}^{2^{D'}} h_1^{(i)}(x_1) \cdot h_2^{(i)}(x_2) \cdot \ldots \cdot h_k^{(i)}(x_k),$$

for some $h_j^{(i)} \colon \{0,1\}^{n/k} \to \{0,1\}$ ($i \in \left[2^{D'}\right], j \in [k]$). Therefore, to show that the PRG G_t δ-fool f, it suffices to show that G_t $\left(\delta/2^{D'}\right)$-fools every function g of the form

$$g(x_1, x_2, \ldots, x_k) = h_1(x_1) \cdot h_2(x_2) \cdot \ldots \cdot h_k(x_k).$$

More specifically we show the following.

Claim 56. *For every $k \geq 2$ and $0 \leq i \leq t$, the generator G_i defined above $\left(3^i \cdot \delta'\right)$-fools every function $g_i \colon \{0,1\}^{n/2^{t-i}} \to \{0,1\}$ of the form*

$$g_i(x_1, x_2, \ldots, x_{k/2^{t-i}}) = h_1(x_1) \cdot h_2(x_2) \cdot \ldots \cdot h_{k/2^{t-i}}(x_{k/2^{t-i}}),$$

where $x_1, x_2, \ldots, x_{k/2^{t-i}} \in \{0,1\}^{n/k}$.

Proof. The proof is by induction on i. The base case is $i = 0$, which is trivial given the definition of G_0. Now suppose the claim holds for $i-1$, we show the case for i. This is done using a hybrid argument. Consider the following four distributions

- $\mathcal{D}_1 = U_{n/2^{t-i}}$,
- $\mathcal{D}_2 = U_{n/2^{t-i+1}} \circ G_{i-1}(U_{r_{i-1}})$,
- $\mathcal{D}_3 = G_{i-1}(U_{r_{i-1}}) \circ G_{i-1}(U'_{r_{i-1}})$ (U and U' are two independent uniform distributions),
- $\mathcal{D}_4 = G_i(U_{r_i})$.

We want show show that

$$|\mathbf{E}[g_i(\mathcal{D}_1)] - \mathbf{E}[g_i(\mathcal{D}_4)]| \leq 3^i \cdot \delta'.$$

By the triangle inequality, it suffices to show that

$$|\mathbf{E}[g_i(\mathcal{D}_1)] - \mathbf{E}[g_i(\mathcal{D}_2)]| + |\mathbf{E}[g_i(\mathcal{D}_2)] - \mathbf{E}[g_i(\mathcal{D}_3)]| + |\mathbf{E}[g_i(\mathcal{D}_3)] - \mathbf{E}[g_i(\mathcal{D}_4)]| \leq 3^i \cdot \delta'. \tag{5}$$

We show Equation (5) by upper bounding each of the three summands.

First summand. We show that

$$|\mathbf{E}[g_i(\mathcal{D}_1)] - \mathbf{E}[g_i(\mathcal{D}_2)]| \leq 3^{i-1} \cdot \delta'. \tag{6}$$

Let us re-write g_i as

$$g_i(x_1, x_2, \ldots, x_{k/2^{t-i}}) = h^{\mathrm{L}}(x_1, x_2, \ldots, x_{k/2^{t-i+1}}) \cdot h^{\mathrm{R}}(x_{k/2^{t-i+1}+1}, x_{k/2^{t-i+1}+2}, \ldots, x_{k/2^{t-i}}),$$

where

$$h^{\mathrm{L}}(y) := \prod_{j=1}^{k/2^{t-i+1}} h_i(y) \quad \text{and} \quad h^{\mathrm{R}}(y) := \prod_{j=k/2^{t-i+1}}^{k/2^{t-1}} h_i(y).$$

Then,

$$\begin{aligned}
\mathbf{E}[g_i(\mathcal{D}_2)] &= \mathbf{E}\left[h^{\mathrm{L}}(U_{n/2^{t-i+1}}) \cdot h^{\mathrm{R}}(G_{i-1}(U_{r_{i-1}}))\right] \\
&= \mathbf{E}\left[h^{\mathrm{L}}(U_{n/2^{t-i+1}})\right] \cdot \mathbf{E}\left[h^{\mathrm{R}}(G_{i-1}(U_{r_{i-1}}))\right] \\
&= \mathbf{E}\left[h^{\mathrm{L}}(U_{n/2^{t-i+1}})\right] \cdot \left(\mathbf{E}\left[h^{\mathrm{R}}(U_{n/2^{t-i+1}})\right] \pm 3^{i-1} \cdot \delta'\right) \\
&\qquad\qquad \text{(By the induction hypothesis)} \\
&= \mathbf{E}\left[h^{\mathrm{L}}(U_{n/2^{t-i+1}})\right] \cdot \mathbf{E}\left[h^{\mathrm{R}}(U_{n/2^{t-i+1}})\right] \pm 3^{i-1} \cdot \delta' \\
&= \mathbf{E}[g_i(\mathcal{D}_1)] \pm 3^{i-1} \cdot \delta',
\end{aligned}$$

as desired.

Second summand. By a similar argument, it can be shown that

$$|\mathbf{E}[g_i(\mathcal{D}_2)] - \mathbf{E}[g_i(\mathcal{D}_3)]| \leq 3^{i-1} \cdot \delta'. \tag{7}$$

We omit the details here.

Third summand. We show that

$$|\mathbf{E}[g_i(\mathcal{D}_3)] - \mathbf{E}[g_i(\mathcal{D}_4)]| \leq \delta'. \tag{8}$$

We have

$$\begin{aligned}
\mathbf{E}[g_i(\mathcal{D}_4)] &= \mathbf{E}[g_i(G_i(U_{r_i}))] \\
&= \mathbf{E}\left[h^{\mathrm{L}}(G_{i-1}(X)) \cdot h^{\mathrm{R}}(G_{i-1}(\mathrm{Ext}_{i-1}(X,Z)))\right] \\
&\qquad\qquad \text{(where } X \sim \{0,1\}^{r_{i-1}} \text{ and } Z \sim \{0,1\}^d\text{)} \\
&= \mathbf{E}[A(X) \cdot B(\mathrm{Ext}_{i-1}(X,Z))] \\
&\qquad \text{(where } A(\cdot) = h^{\mathrm{L}}(G_{i-1}(\cdot)) \text{ and } B(\cdot) = h^{\mathrm{R}}(G_{i-1}(\cdot))\text{)} \\
&= \mathbf{E}[B(\mathrm{Ext}_{i-1}(X,Z)) \mid A(X)=1] \cdot \mathbf{Pr}[A(X)=1].
\end{aligned}$$

Similarly, we get

$$\mathbf{E}[g_i(\mathcal{D}_3)] = \mathbf{E}[B(U_{r_{i-1}}) \mid A(X)=1] \cdot \mathbf{Pr}[A(X)=1].$$

As a result, we have

$$\begin{aligned}
&|\mathbf{E}[g_i(\mathcal{D}_4)] - \mathbf{E}[g_i(\mathcal{D}_3)]| \\
&= \left|\left(\mathbf{E}[B(\mathrm{Ext}_{i-1}(X,Z)) \mid A(X)=1] - \mathbf{E}[B(U_{r_{i-1}}) \mid A(X)=1]\right) \cdot \mathbf{Pr}[A(X)=1]\right|.
\end{aligned} \tag{9}$$

On the one hand, if $\mathbf{Pr}[A(X)=1] \leq \delta'$, then Equation (9) is at most δ'. On the other hand, if $\mathbf{Pr}[A(X)=1] > \delta'$, then

$$H_\infty(X \mid A(X)=1) > r_{i-1} - \log(1/\delta') > r_{i-1} - D' - 2t - \log(1/\delta) = \kappa_{i-1}.$$

Then by the fact that Ext_{i-1} is a (κ_{i-1}, δ')-extractor, we have

$$\left|\mathbf{E}[B(\mathrm{Ext}_{i-1}(X,Z)) \mid A(X)=1] - \mathbf{E}[B(U_{r_{i-1}}) \mid A(X)=1]\right| \leq \delta'.$$

Therefore, Equation (9) is at most δ' and this complete the proof of Equation (8). Finally, note that Equation (5) follows from Equation (6), Equation (7) and Equation (8). This completes the proof of Theorem 56. □ (Theorem 56)

Given Theorem 56, Theorem 28 now follows by letting $i = t$. □ (Theorem 28)

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